Following the Information Bound from a Search-Window Prior with Sequential Bayesian Experiment Design in Nitrogen-Vacancy Magnetometry

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(Dated: November 3, 2026)

[Written last, once the simulation and lab results are in.]

I. INTRODUCTION

The negatively charged nitrogen-vacancy (N-V) centre in diamond is a solid-state electron spin that can be initialised and read out optically at room temperature, which makes it a magnetic field sensor with nanoscale spatial resolution [1–3]. In optically detected magnetic resonance (ODMR) magnetometry the field to be measured shifts the spin resonance, and it is read from the position of that resonance, found by recording the photoluminescence at a sequence of microwave drive frequencies within a search window.

Choosing these drive frequencies well is an active field. Non-adaptive undersampling measures only a subset of the window and reconstructs the line from it, either by fitting a Lorentzian to randomly chosen drive frequencies [4] or by compressed sensing in a sparse spectral basis [5]. Bayesian experiment design instead keeps a posterior over the parameters, here as a sequential Monte Carlo (SMC) particle cloud, and chooses each next measurement to maximise its expected utility, such as the information it is expected to gain [6]. In N-V Ramsey magnetometry, adaptive Bayesian schemes set the readout phase or the precession time from the outcomes so far [7–9]; in ODMR, the same approach chooses the next drive frequency and reaches a given precision more than an order of magnitude faster than a sweep [10, 11].

In recent years, learning the design rule itself has become a particularly active direction. Early work learned policies for adaptive phase estimation [12, 13]. Reinforcement learning has since been used to find the controls that make a quantum sensor most sensitive [14–16] and to estimate several parameters at once in a photonic experiment [17]. Neural networks trained by an evolution strategy or reinforcement learning replace the costly expected-gain optimisation of Bayesian design by a fast learned heuristic [18], and model-aware reinforcement learning trains such a policy through a particle-filter posterior, with applications to N-V Ramsey magnetometry [19, 20]. For a wide field range, a Bayesian neural network first narrows the range and a reinforcement-learning agent then refines the estimate [21].

What is missing from this research is a reference for the ODMR search itself: a bound on how fast any strategy can locate the resonance. Without it, the reported speedups cannot be placed against the best achievable, it is not known how many of the measurements are unavoidable, and the observed growth of the speedup as the line occupies less of the window [10] remains unexplained.

Estimation theory provides such references. The Cramér–Rao bound limits the variance of an unbiased estimate by the Fisher information of the data [22], but it ignores what is known before the first measurement. The van Trees inequality, a Bayesian form of the Cramér–Rao bound, adds the information of the prior and holds for any estimator [23, 24]. A search window is exactly such prior knowledge, so this bound is a natural yardstick for an ODMR search, and the information it counts is the same quantity that Bayesian experiment design maximises. In this work we use it to analyse sequential Bayesian experiment design (SBED) and drive-frequency sweeps for CW and pulsed ODMR.

Section II derives the bound from the physical model and the speedup, sensitivity and bandwidth that follow from it; Sec. III describes the particle filter, the measurement strategies and how they are evaluated; Sec. IV presents the simulations and the experiment; and Sec. V discusses the limits of the approach.

II. THEORY

A. Physical model

The negatively charged N-V centre is described by five levels [3, 25] (Fig. 1): the spin-triplet ground state $^3A_2$, split into $|0\rangle_g$ and $|\pm1\rangle_g$ by the spin projection m_s on the N-V axis; the spin-triplet excited state ^3E, with $|0\rangle_e$ and $|\pm1\rangle_e$; and a metastable spin singlet $|s\rangle$ that stands for the singlet manifold. Green light drives the spin-conserving optical transitions $|m_s\rangle_g \to |m_s\rangle_e$, which decay back by emitting red photoluminescence (PL). From $|\pm1\rangle_e$, however, the intersystem crossing to the singlet is much faster than from $|0\rangle_e$, and the singlet decays preferentially to $|0\rangle_g$. Two consequences follow. Under continuous illumination the spin is pumped into $m_s = 0$; and a centre in $m_s = \pm1$ spends part of its optical cycles in the dark singlet, so it emits less PL than one in $m_s = 0$. The spin state can therefore be prepared and read out optically.

With a static field B_z along the N-V axis, the ground-state spin ($S = 1$) Hamiltonian is, in units with $h = 1$ (energies in frequency units) [3],

$$H = D\,S_z^2 + \gamma_e B_z S_z + A_\parallel S_z I_z, \tag{1}$$

with $D \approx 2.87$ GHz the zero-field splitting, $\gamma_e \approx$

[Level diagram: $^3A_2$ and ^3E triplets ($m_s = 0, \pm 1$), singlet $|s\rangle$; green pumping, red PL, spin-selective intersystem crossing; microwave transitions at $D \pm \Delta_Z$.]

FIG. 1. Five-level model of the N-V centre [3, 25]. [Caption to follow the figure.]

[Sequence diagram: CW ODMR (laser and microwaves on together, PL read continuously) and pulsed ODMR (laser pulse to polarise, microwave π-pulse of length $1/(2\Omega)$ with the laser off, laser pulse to read out).]

FIG. 2. Laser and microwave sequences of CW and pulsed ODMR. [Caption to follow the figure.]

28 GHz/T the electron gyromagnetic ratio, and $A_\parallel$ the axial hyperfine coupling to the nitrogen nuclear spin I ($|A_\parallel| = 2.15$ MHz for ^{14}N, $I = 1$, and 3.03 MHz for ^{15}N, $I = 1/2$ [26]). The strain splitting $E(S_x^2 - S_y^2)$ is dropped, a small correction when $\gamma_e B_z \gg E$. The eigenstates are the product states $|m_s, m_I\rangle$, and the magnetic-dipole transitions $m_s = 0 \to \pm 1$, which conserve m_I, lie at

$$f_\pm(m_I) = D \pm \left(\Delta_Z + A_\parallel m_I\right), \qquad \Delta_Z = \gamma_e B_z, \tag{2}$$

where Δ_Z is the Zeeman shift. Each electron transition therefore splits into $2I+1$ hyperfine lines spaced by $|A_\parallel|$: a triplet for ^{14}N and a doublet for ^{15}N.

An ODMR measurement sets the microwave drive to a frequency ν and records the PL, normalised to its off-resonant level. When ν meets f_+ or f_-, the drive transfers population from $m_s = 0$ to the darker $m_s = \pm 1$ state and the PL dips. D is treated as a known constant, so the position f_+ of the upper line carries the field, $f_+ - D = \Delta_Z$, and an uncertainty in f_+ is an uncertainty in Δ_Z.

Near the excited-state level anticrossing the nitrogen nuclear spin is polarised into a single m_I, and each transition shows a single line [27]; its fixed offset $A_\parallel m_I$ is absorbed into f_+. The same single line describes a line whose hyperfine structure is unresolved, $\Gamma \gg |A_\parallel|$. The resolved triplet and doublet are treated at the end of this section. A drive near f_+ couples only $|0\rangle_g$ and $|{+1}\rangle_g$, and the excited state and the singlet enter only through the rates at which the laser repolarises the spin and destroys its coherence [28], so the spin is a driven two-level system $\{m_s = 0, m_s = +1\}$. The laser prepares it in $m_s = 0$; a microwave drive of Rabi frequency Ω (Hz) at detuning $\Delta = \nu - f_+$ then transfers it to $m_s = +1$ after a time t with probability

$$P(\Delta, t) = \frac{\Omega^2}{\Omega^2 + \Delta^2} \sin^2\!\left(\pi t \sqrt{\Omega^2 + \Delta^2}\right), \tag{3}$$

and the PL dip is proportional to P. The line shape depends on how the drive is applied (Fig. 2).

CW ODMR. Laser and microwaves are on together, and the spin is driven for a time T much longer than one Rabi period, $T\sqrt{\Omega^2 + \Delta^2} \gg 1$, while the laser keeps resetting the phase of the oscillation. Only the time average of Eq. (3) is then seen, and since $\frac{1}{T}\int_0^T \sin^2\!\left(\pi t\sqrt{\Omega^2+\Delta^2}\right) dt \to \frac{1}{2}$ in this limit,

$$\bar{P}(\Delta) = \frac{1}{2}\,\frac{\Omega^2}{\Omega^2 + \Delta^2}, \tag{4}$$

a Lorentzian in Δ of half width Ω. This is the microwave contribution to the width, power broadening: the stronger the drive, the further off resonance it still flips the spin.

Eq. (4) treats the drive alone. In CW ODMR the laser acts on the spin at the same time, in two ways: each optical cycle pumps it back towards $m_s = 0$, at a rate Γ_p, and destroys the phase between $m_s = 0$ and $m_s = +1$, at a rate Γ_c. These are a longitudinal and a transverse relaxation rate, $1/T_1$ and $1/T_2$ of the two-level system. The steady state of the Bloch equations with these rates and a drive of angular Rabi frequency $\Omega_R = 2\pi\Omega$ is [28]

$$P_{\rm ss}(\Delta) = \frac{1}{2}\,\frac{\Omega_R^2\,\Gamma_c/\Gamma_p}{(2\pi\Delta)^2 + \Gamma_c^2 + \Omega_R^2\,\Gamma_c/\Gamma_p}, \tag{5}$$

which reduces to Eq. (4) when the two rates are equal. It is still a Lorentzian, now of half width

$$\gamma = \frac{1}{2\pi}\sqrt{\Gamma_c^2 + \Omega_R^2\,\frac{\Gamma_c}{\Gamma_p}}. \tag{6}$$

The two terms add in quadrature. The first is the optical broadening: the width that remains at vanishing microwave power, set by how fast the laser destroys the spin's coherence. The second is the microwave power broadening of Eq. (4), scaled by Γ_c/Γ_p. Both rates grow with the optical saturation parameter s as $\Gamma_{p,c} = \Gamma_{p,c}^\infty\, s/(1+s)$, so the line is set by the laser power at weak microwave drive and by the microwave power at strong drive. The depth, the value of Eq. (5) at $\Delta = 0$, depends on both powers as well [28]. Whatever its origin, the line is one Lorentzian dip,

$$S(\nu) = 1 - C\,\frac{\gamma^2}{\gamma^2 + (\nu - f_+)^2}, \tag{7}$$

with contrast C (the depth of the dip), half width γ of Eq. (6) and full width at half maximum $\Gamma = 2\gamma$. Optical and microwave broadening are not modelled separately: they enter only through the single linewidth Γ, which is inferred with the line. Eq. (7) is the line used in Sec. II B.

Pulsed ODMR is used to eliminate this power broadening [28]. In CW ODMR, Eq. (6), the laser that reads the spin out also broadens the line through Γ_c, and narrowing it means lowering both powers and with them the contrast and the PL. Pulsed ODMR separates the two in time: the laser first polarises the spin and is then switched off while a single microwave π-pulse of duration $t_\pi = 1/(2\Omega)$ is applied, after which a second laser pulse reads the spin out (Fig. 2). The laser therefore no

longer broadens the line, the microwave width is set by the pulse length rather than by a steady drive power, and Eq. (3) is evaluated at the one fixed time t_π instead of being averaged. With $r = \sqrt{1 + \Delta^2/\Omega^2}$,

$$P_\pi(\Delta) = \frac{\sin^2(\pi r/2)}{r^2}, \tag{8}$$

which is not a Lorentzian but a sinc^2-like line: for $|\Delta| \gg \Omega$, $r \approx |\Delta|/\Omega$ and $P_\pi \approx \frac{\pi^2}{4}\,\mathrm{sinc}^2(\pi\Delta t_\pi)$, with $\mathrm{sinc}\,u = \sin u/u$: the power spectrum of a rectangular pulse of length t_π. Its features follow from Eq. (8). On resonance $r = 1$ and the inversion is complete, $P_\pi = 1$; the half width is $0.80\,\Omega$, so the line narrows as the pulse gets longer, the Fourier limit $\propto 1/t_\pi$. The line vanishes wherever $\sin(\pi r/2) = 0$, i.e. $r = 2, 4, \ldots$, at $|\Delta| = \sqrt{3}\,\Omega, \sqrt{15}\,\Omega, \ldots$: there the spin completes a whole number of full Rabi cycles and returns to $m_s = 0$. Between the zeros lie side lobes that decay as $1/r^2$; the first, near $r = 3$ at $|\Delta| \approx 2.7\,\Omega$, is 12% of the depth. Side lobes are a hazard for a search: a lobe is a dip where there is no resonance.

T_2^* sets how far this narrowing can go. The spin's transition frequency is not sharp but fluctuates from run to run (mostly through the ^{13}C nuclear-spin bath), a Gaussian spread of width $\sigma_\delta = 1/(\sqrt{2}\,\pi T_2^*)$, so the measured line is Eq. (8) averaged over that spread,

$$\bar{P}_\pi(\Delta) = \int P_\pi(\Delta - \delta)\,\frac{e^{-\delta^2/2\sigma_\delta^2}}{\sqrt{2\pi}\,\sigma_\delta}\,d\delta. \tag{9}$$

For $t_\pi \ll T_2^*$ the spread is small against Ω and the line is the sinc^2-like Eq. (8). For $t_\pi \gtrsim T_2^*$ the Gaussian dominates: the zeros and side lobes are washed out, the line stops narrowing at a width $\sim 1/T_2^*$, and the depth falls because the pulse no longer inverts every frequency class [28]. The best pulsed line is therefore reached at $t_\pi \sim T_2^*$, a width set by the spin itself rather than by the laser or microwave power. CW ODMR and its Lorentzian line are the main case of this work, and the information analysis of Sec. II B is derived for it; pulsed ODMR is treated as well, with its own line shape, Eq. (9), in place of Eq. (7). [Where pulsed goes (a Theory subsection for the information of the pulsed line, and its own Results).]

The spectrum is mirror symmetric about D, so the drive window is placed above D and holds the upper transition only. The tail of the lower one, at $D - \Delta_Z$, adds $C\gamma^2/[\gamma^2 + (\nu - D + \Delta_Z)^2]$ inside the window: at the line centre a fraction $\gamma^2/(\gamma^2 + 4\Delta_Z^2)$ of C, i.e. 20% at $\Delta_Z = \gamma$ and below 1% for $\Delta_Z \gtrsim 5\gamma$. Equation (7) assumes $\Delta_Z \gg \gamma$. Small Zeeman shifts, where the tails of the two lines overlap, are a separate problem and outside the scope of this work.

f_+ is the parameter of interest. The width Γ and the contrast C are nuisance parameters: they are inferred jointly with f_+ but only matter through their effect on it. The noise level σ (below) is inferred as well.

One shot is a PL reading at a fixed drive frequency. A shot that collects N_{ph} photons on average has Poisson-distributed counts, so the normalised reading has a standard deviation of at least $N_{\mathrm{ph}}^{-1/2}$ (to first order in C): the photon shot-noise limit, reached when technical noise (laser intensity, microwave power, collection drift) is negligible. For $N_{\mathrm{ph}} \gg 1$ the counts are close to Gaussian, and the single-shot reading is

$$y = S(\nu) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \qquad \sigma \geq N_{\mathrm{ph}}^{-1/2}. \tag{10}$$

A measurement is the mean of k shots at the same ν. Its noise is $\sigma/\sqrt{k}$, so its Fisher information, Eq. (13), grows linearly with k. At the shot-noise limit it grows with the total number of photons kN_{ph} collected at that drive frequency, which no averaging scheme can beat. The signal-to-noise ratio of a single shot depends on where it is taken: it is the drop in PL at ν over the noise,

$$\mathrm{SNR}(\nu) = \frac{1 - S(\nu)}{\sigma} = \frac{C}{\sigma}\,\frac{\gamma^2}{\gamma^2 + (\nu - f_+)^2}. \tag{11}$$

It is largest on resonance, $\nu = f_+$, where

$$\mathrm{SNR} \equiv \mathrm{SNR}(f_+) = C/\sigma, \tag{12}$$

the single number by which a line is labelled. The information of a measurement, however, follows the local slope of S rather than its depth, Eq. (13): it vanishes on resonance and peaks at $|\nu - f_+| = \gamma/\sqrt{3}$, where the dip is 3/4 deep, which gives Eq. (14). Equation (12) also neglects the mirror line: its tail raises the on-resonance depth to $C[1 + \gamma^2/(\gamma^2 + 4\Delta_Z^2)]$, so C/σ is the $\Delta_Z \gg \gamma$ limit of the on-resonance SNR. In the simulations each measurement is one Gaussian draw, i.e. $k = 1$, and σ is set from the SNR. All SNR values in this paper follow Eq. (12).

The prior knowledge of the experimentalist is an interval of width W that holds the resonance, $f_+ \in [f_{\min}, f_{\min} + W]$; W is the prior support of f_+ that enters Eq. (15), and the drive frequency is scanned over at least this interval. The ratio W/Γ, the inverse of the fraction of the window the line occupies, sets how hard the search is. [Code fix in progress: the particle filter's prior on Δ_Z becomes uniform over the range the truth is drawn from (it is now an offset Gaussian plus 10% uniform); the nuisance priors stay offset on purpose. Then state W of the simulations here: is it the drive window W_{d} or the Δ_Z range?]

The code also models a pseudo-Voigt line whose total width is the linear sum $\Gamma = \Gamma_L + \Gamma_G$ of its Lorentzian and Gaussian parts (the study varies the Gaussian share at fixed Γ), and the resolved hyperfine lines of Eq. (2), a ^{14}N triplet or a ^{15}N doublet whose lines share the same total contrast; the main text keeps the single Lorentzian of Eq. (7). [Where the hyperfine-resolved runs are used, and how f_+ is defined for them (centre line or m_I-averaged).]

B. Fisher information of a single measurement

From here on, the statistics are written in the common notation of Bayesian experimental design: the design $x \equiv$

ν is the drive frequency, $\theta \equiv f_+$ the parameter of interest, $\phi = (\Gamma, C)$ the nuisance parameters, y the outcome of a measurement, and $S(x;\theta,\phi)$ the line of Eq. (7). One measurement at x, the mean of k shots with single-shot noise σ, carries Fisher information about θ

$$I(x) = \frac{k\, g(x)^2}{\sigma^2}, \qquad g(x) = \frac{\partial S(x;\theta,\phi)}{\partial \theta}. \tag{13}$$

For a Lorentzian dip of depth C and full width Γ, $S = 1 - C\,\gamma^2/(\gamma^2+u^2)$ with $\gamma = \Gamma/2$ and $u = x-\theta$, the slope is steepest at $|u| = \gamma/\sqrt{3}$, where $|g^*| = 9C/(4\sqrt{3}\,\Gamma) \approx 1.30\,C/\Gamma$. The best possible information per measurement is therefore

$$I^* = \max_x I(x) = \frac{27}{16}\, k\, \frac{\mathrm{SNR}^2}{\Gamma^2}, \qquad \mathrm{SNR} = C/\sigma. \tag{14}$$

I^* assumes the nuisance parameters (width, contrast) are known; with free nuisances it is an upper bound on the usable information. The nuisances are inferred along with θ but are not the subject of this work.

C. Prior information: the search window

Information about θ is measured in inverse variance, $J = \sigma_\theta^{-2}$ (Hz^{-2}); independent measurements add their $I(x)$ to it. The same currency measures what is known *before* the first measurement. If the experimentalist knows only that the resonance lies somewhere in a window of width W, the state of knowledge is a uniform prior $p(\theta) = 1/W$, whose variance is

$$\sigma_0^2 = \int_{-W/2}^{W/2} \frac{u^2}{W}\, du = \frac{W^2}{12}, \qquad J_0 \equiv \sigma_0^{-2} = \frac{12}{W^2}. \tag{15}$$

The search window is therefore a prior: it holds J_0 of information that no measurement had to pay for, and a narrower window starts with more. (A uniform prior has no finite Fisher information at its edges, so the inverse variance is used as the convention.) The unitless gain after the measurements is $\log_2(\sigma_0/\sigma)$ bits.

D. Van Trees bound

Because the prior carries information, the relevant bound is not the Cramér–Rao bound, which holds for unbiased estimators and ignores J_0, but the van Trees inequality [23, 24]. It holds for any estimator, averaged over the prior: measurements are independent given θ, so their information adds to the prior's,

$$\sigma_\theta(N) \;\geq\; \Big(J_0 + \sum_{n=1}^{N} I(x_n)\Big)^{-1/2} \;\geq\; \sigma_B(N) = \Big(\frac{12}{W^2} + N I^*\Big)^{-1/2}. \tag{16}$$

The search, focus and refine regimes of a run are the limits of this bound and the crossover between them.

a. Prior limit (search). For $\sum_n I(x_n) \ll J_0$, $\sigma \to \sigma_0 = W/\sqrt{12}$: the knowledge is still that of the window. The estimate leans on the prior and is biased towards its mean, which is why it may lie below the unbiased Cramér–Rao bound.

b. Data limit (refine). For $\sum_n I(x_n) \gg J_0$, the prior is washed out and the posterior mean becomes asymptotically unbiased and efficient (Bernstein–von Mises [29]): σ_B tends to the unbiased Cramér–Rao bound $(NI^*)^{-1/2}$ [22], a slope $-\frac{1}{2}$ reached only by measuring at the steepest slope.

c. Crossover (focus). With I^* at every step the crossover would come at $N_c = J_0/I^* = \frac{64}{9}\,(\Gamma/W)^2/(k\,\mathrm{SNR}^2)$, far below one measurement for any realistic window ($N_c \approx 2\times10^{-4}$ for $W = 20\,\Gamma$ at $\mathrm{SNR} = 10$). It comes much later because I^* is reached only once the dip has been found: a measurement on the baseline carries $I(x) \approx 0$, so the sum in Eq. (16) stays near J_0 while the strategy searches. What such a measurement contributes instead is *elimination*: it shrinks the prior itself.

d. Elimination model. Consider a sweep with spacing Δ (e.g. $\Delta = \Gamma/3$) at an SNR high enough that one measurement inside the dip reveals it. After n measurements without a dip, the posterior is uniform on the part of the window not yet excluded, so

$$\sigma(n) = \frac{W - n\Delta}{\sqrt{12}}, \qquad n < n_d, \tag{17}$$

where $n_d \leq W/\Delta$ is the step that finds the dip (on average $W/(2\Delta)$ over a uniformly placed resonance). In information terms $J(n) = 12/(W-n\Delta)^2$: the gain per step is not constant but grows as the remaining width shrinks. Detection localises the resonance to about one grid step, after which the measurements add Fisher information:

$$\sigma(n) = \Big(\frac{12}{\Delta^2} + (n-n_d)\, I^*\Big)^{-1/2}, \qquad n \geq n_d. \tag{18}$$

Eqs. (17)–(18) reproduce both limits: $\sigma \to \sigma_0$ as $n \to 0$ and $\sigma \to (nI^*)^{-1/2}$ for $n \gg n_d$, the two limits of Eq. (16). For a wide window the three regimes become visible (Fig. 3): the prior limit is a plateau, since Eq. (17) falls linearly and on a log–log axis looks flat until about half the window is excluded, for a number of steps that grows with W/Δ; the crossover is a cliff steeper than $-\frac{1}{2}$, where detection collapses σ from about the window to about Δ (roughly $\log_2(W/\Delta)$ bits) and which ends when the strategy can measure at the steepest slope, $\sigma \approx \varepsilon \approx 0.075\,\Gamma$; the data limit is the $-\frac{1}{2}$ tail. A narrow window ($W \lesssim 3.5\,\Gamma$, $\sigma_0 \lesssim \Gamma$) has no plateau. Search, focus and refine are therefore not separate mechanisms but the regimes of one bound.

The elimination model only illustrates the shape of the profile. At low SNR one measurement no longer reveals the dip and n_d grows, but Eq. (16) holds at any SNR and needs no detection model; the runs are read against

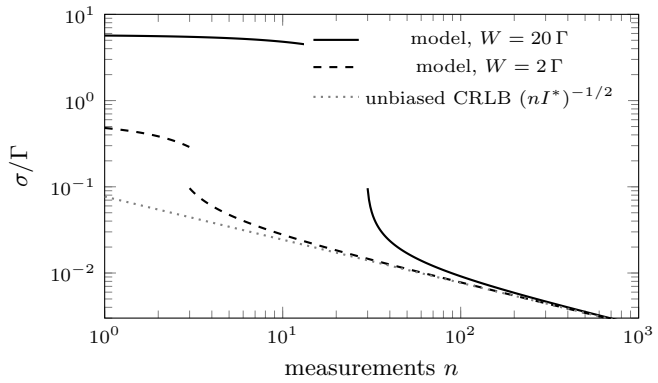


FIG. 3. Elimination model, Eqs. (17)–(18), at illustrative values (SNR 10, $k = 1$, $\Delta = \Gamma/3$, dip found halfway through the grid). The wide window shows a search plateau, a detection cliff and the $-\frac{1}{2}$ refine tail; the narrow window shows no plateau. Both start at the prior limit σ_0 of the van Trees bound and tend to its data limit, the unbiased Cramér–Rao bound, at large n. [Schematic or drawn with the study's grid values?]

it alone. The false-alarm test of Sec. III B only steers where the strategy measures.

E. Expected information gain and the bound

The expected information gain of measuring at x is the mutual information between the parameters and the outcome, i.e. the expected Kullback–Leibler divergence from prior to posterior [30]:

$$\begin{aligned}\mathrm{EIG}(x) &= \mathbb{E}_{y|x}\Big[D_{\mathrm{KL}}\big(p(\theta,\phi \mid y,x)\,\|\,p(\theta,\phi)\big)\Big] = I(\theta,\phi;y \mid x)\\ &= H[y \mid x] - \mathbb{E}_{\theta,\phi}H[y \mid \theta,\phi,x]. \qquad (19)\end{aligned}$$

The noise is modelled as Gaussian with the same variance σ^2/k at every drive frequency (Sec. II A); this is an assumption of the model, not a limit, since otherwise every drive frequency would need its own noise model. Hence, for a known σ, $H[y \mid \theta,\phi,x] = \frac{1}{2}\ln(2\pi e\,\sigma^2/k)$; the particle filter, which learns σ (Sec. III A), uses its posterior mean. For a given variance the Gaussian has the largest entropy [31], so $H[y \mid x] \le \frac{1}{2}\ln 2\pi e\,(\sigma^2_{\mathrm{pred}}(x) + \sigma^2/k)$, with $\sigma^2_{\mathrm{pred}}(x)$ the variance of $S(x;\theta,\phi)$ over the posterior. Hence

$$\mathrm{EIG}(x) \le \tfrac{1}{2}\ln\Big(1 + \frac{k\,\sigma^2_{\mathrm{pred}}(x)}{\sigma^2}\Big), \qquad (20)$$

which is what SBED (Sec. III B) evaluates on its particles. Equality holds when the predictive distribution of $S(x;\theta,\phi)$ is Gaussian.

a. When the variance is a good proxy: the local-linear regime. The only remaining gap in Eq. (20) is the shape of the posterior. By the Bernstein–von Mises theorem, the Bayesian central limit theorem [29], the posterior tends with growing data to a normal distribution centred near the truth with variance $1/J_n$. A Gaussian's entropy depends on its variance alone, $H[\theta] = \frac{1}{2}\ln(2\pi e\,\sigma^2_\theta)$, so the expected KL divergence of one step reduces to a ratio of variances, $\frac{1}{2}\ln(\sigma^2_{\theta,n}/\sigma^2_{\theta,n+1})$. Once the posterior is also narrow compared with the line, S is linear over it, the predictive distribution of $S(x;\theta,\phi)$ is Gaussian, Eq. (20) holds with equality, $\sigma^2_{\mathrm{pred}}(x) \approx g(x)^2\sigma^2_\theta$, and Eq. (20) becomes

$$\mathrm{EIG}(x) \approx \tfrac{1}{2}\ln\big(1 + I(x)\,\sigma^2_\theta\big) = \tfrac{1}{2}\ln\frac{J_n + I(x)}{J_n}. \qquad (21)$$

Greedy EIG then maximises $I(x)$, i.e. measures at the steepest slope, and the gains telescope: $\sum_n \mathrm{EIG}_n = \frac{1}{2}\ln(J_N/J_0) = \ln(\sigma_0/\sigma_N)$. The acquisition rule and the information budget of Sec. II D are the same quantity. The optimal strategy, which would measure every point at the steepest slope of the true line, reaches the data limit of Eq. (16); greedy EIG approaches it once the posterior is narrow compared with the line, while during the search it is only myopically optimal, which is where the strategies of Sec. III B differ.

b. When it is not: search. Before the dip is found, the predictive distribution at x is two-valued: with probability p (the posterior mass that would put a dip at x) the signal is $1-C$, otherwise 1. Then $\sigma^2_{\mathrm{pred}} = p(1-p)C^2$, while the true mutual information is at most the binary entropy $H_b(p)$. The variance form keeps growing with SNR where the true gain saturates, so it overvalues search candidates relative to refine candidates; both peak at $p = \frac{1}{2}$, so they rank search candidates alike. [Mention computing the exact mixture entropy from the particles, or leave out?]

F. Speedup over a frequency sweep

The same bound predicts the speedup over a conventional sweep. A sweep spreads its measurements uniformly over the window, so its information per measurement is the window average of Eq. (13),

$$\bar I = \frac{1}{W}\int I(x)\,dx = \frac{k}{\sigma^2 W}\int g(x)^2\,dx = \frac{\pi}{4}\,\frac{k\,C^2}{\sigma^2\,\gamma\,W} \qquad (22)$$

for a Lorentzian ($\int(\partial_\theta L)^2 du = \pi/4\gamma$), valid for $W \gg \Gamma$. The sweep therefore accumulates information at the rate $\bar I$ from its first measurement, $\sigma \approx (J_0 + N\bar I)^{-1/2}$, with no plateau and no cliff, as observed by Dushenko *et al.* [10]. An adaptive strategy pays the search cost n_d first and then gains I^* per measurement. The number of measurements to reach a target τ is

$$N_{\mathrm{sweep}}(\tau) = \frac{\tau^{-2} - J_0}{\bar I}, \qquad N_{\mathrm{ad}}(\tau) = n_d + \frac{\tau^{-2} - 12/\Delta^2}{I^*}, \qquad (23)$$

and the speedup is $S(\tau) = N_{\mathrm{sweep}}/N_{\mathrm{ad}}$. For a precise target the refine terms dominate and

$$S_\infty = \frac{I^*}{\bar{I}} = \frac{27}{16\pi}\frac{W}{\gamma} = \frac{27}{8\pi}\frac{W}{\Gamma} \approx 1.07\,\frac{W}{\Gamma}. \tag{24}$$

The speedup is set by the inverse occupancy W/Γ of the window, which Dushenko *et al.* [10] identified qualitatively; Eq. (24) gives its prefactor. It is independent of the SNR, because both strategies gain information $\propto \mathrm{SNR}^2$. For a loose target the search cost n_d dominates N_{ad} and $S(\tau) < S_\infty$, so the full dependence on window, linewidth, SNR (through n_d at low SNR) and target is given by Eq. (23). A sweep cannot approach the van Trees bound at any N: its rate $\bar{I}$ is fixed by the window, while the bound's data limit requires every measurement at the steepest slope. [Check Eq. (24) against the speedup reported in Ref. [10] for their window and line structure (a resolved triplet, not one Lorentzian).]

G. Sensitivity and bandwidth

Sensitivity and bandwidth follow from the same budget once measurements are converted to time. Let one measurement take $t_m = k\,t_s$ (k shots of duration t_s), so that $T = Nt_m$. With D known, $f_+ - D = \Delta_Z = \gamma_e B$ along the N-V axis ($\gamma_e \approx 28$ GHz/T), so $\sigma_B = \sigma_{f_+}/\gamma_e$.

a. Sensitivity (refine). In the refine regime $\sigma_{f_+} = (NI^*)^{-1/2}$, so $\sigma_{f_+}\sqrt{T}$ is a constant of the experiment:

$$\eta = \frac{\sigma_{f_+}\sqrt{T}}{\gamma_e} = \frac{1}{\gamma_e}\sqrt{\frac{t_m}{I^*}} = \frac{4}{3\sqrt{3}}\,\frac{\Gamma\sqrt{t_s}}{\gamma_e\,\mathrm{SNR}_1}, \tag{25}$$

with $\mathrm{SNR}_1 = C/\sigma$ the single-shot SNR. The prefactor $4/(3\sqrt{3})$ is that of the standard CW-ODMR sensitivity of a Lorentzian line [28]: the familiar formula is the refine limit of the budget, reached only by a strategy that measures at the steepest slope. A strategy whose refine measurements carry on average a fraction $\bar{\eta}_F$ of I^* reaches $\eta/\sqrt{\bar{\eta}_F}$.

b. Bandwidth (whole run). A field reading is useful once the error reaches a target τ; the time needed, $T(\tau)$, sets the bandwidth $1/T(\tau)$. From the elimination model,

$$T(\tau) = t_m\Big[n_d + \frac{\tau^{-2} - 12/\Delta^2}{I^*}\Big], \qquad \eta_{\mathrm{eff}}(\tau) = \frac{\tau\sqrt{T(\tau)}}{\gamma_e}. \tag{26}$$

For $\tau \to 0$ the refine term dominates and $\eta_{\mathrm{eff}} \to \eta$ of Eq. (25). For a loose target the search term dominates: the bandwidth cannot exceed about $1/(n_d t_m) \sim \Delta/(Wt_m)$, whatever the SNR, and η_{eff} is worse than η. The window, i.e. the prior, therefore limits the bandwidth, while the line shape and SNR limit the sensitivity. [Per-shot time is not yet in the simulated data, so results are in measurements; take t_s from the lab.]

III. METHODS

A. Bayesian inference

The posterior is a weighted particle cloud $\{(\theta_i, \phi_i), w_i\}_{i=1}^{N_p}$, $N_p = 1000$, updated by sequential Monte Carlo [32]. Each particle holds the parameter of interest $\theta_i = D + \Delta_{Z,i}$, through its Zeeman shift (D is fixed), and the nuisances $\phi_i = (\Gamma_L, \Gamma_G, C)_i$ of Sec. II A, in the notation of Sec. II B. Every dimension is scaled to $[0, 1]$ by its prior bounds, so one resampling kernel serves all of them; reported values are mapped back to Hz. The noise level is learned but is not a particle dimension: each particle carries a conjugate inverse-gamma posterior on σ^2, with shape α shared by all particles and scale β_i built from that particle's residuals, both discounted by 0.99 per measurement. Integrating σ^2 out, a measurement y_n (mean of k shots at x_n) reweights the particles by a Student-t predictive,

$$w_i \;\propto\; w_i\; t_{2\alpha}\Big(y_n;\, S(x_n;\theta_i,\phi_i),\, \frac{\beta_i}{k\alpha}\Big), \tag{27}$$

so the noise is Gaussian in the model and only its level is unknown. When the effective sample size $(\sum_i w_i^2)^{-1}$ falls below $0.2\,N_p$, the cloud is resampled systematically [33] and rejuvenated with the Liu–West kernel [34],

$$\psi_i' \sim \mathcal{N}\big(a\psi_i + (1-a)\bar{\psi},\; (1-a^2)\Sigma\big), \qquad a = 0.98, \tag{28}$$

with $\psi_i = (\theta_i, \phi_i)$ the whole particle, $\bar{\psi}$, Σ the weighted mean and covariance before resampling, which the kernel preserves. Two safeguards depart from it: the diagonal of Σ is floored at $(0.01\,e^{-n/25})^2$ of each unit range, so early rejuvenations can widen the cloud, and particles leaving the cube are reflected back. Each β_i is then recomputed from the full history at the particle's new position. The full algorithm is in the Supplemental Material [SM]. The initial cloud is drawn from the prior: in the simulations a truncated Gaussian on each dimension [align with Sec. II C].

The particle filter reports the posterior mean and the weighted particle standard deviation,

$$\hat{\theta}_N = \sum_i w_i\theta_i, \qquad \hat{\sigma}_N = \Big[\sum_i w_i(\theta_i - \hat{\theta}_N)^2\Big]^{1/2}, \tag{29}$$

the claimed uncertainty whose meaning Sec. III C discusses.

B. Measurement strategies

Every strategy uses the particle filter of Sec. III A, with the same prior, the same budget of 2500 measurements and the same stop rule, so $\hat{\sigma}_N$ is the same quantity for all of them; the strategies differ only in how the next drive frequency x is chosen. The run stops when the claimed uncertainty of every parameter has been below

its ceiling (100 kHz for θ, 500 kHz for the widths, 0.01 for the contrast) for 8 consecutive measurements and the dip test below finds a dip.

a. Dip test. The test asks where the measurements, not the particles, show a dip. Each measurement gets a score $z_n = (b - y_n)/s$ against a baseline b (the 70th percentile of the y_n, corrected for noise), with s and the Student-t degrees of freedom from the noise posterior of Sec. III A. Every contiguous stretch of measured drive frequencies whose span lies in the prior range of the linewidth gets the boxcar score $\sum z_n/\sqrt{n}$; a stretch is significant when its score exceeds the Bonferroni threshold for a family-wise false-alarm probability $p_{\mathrm{FA}} = 0.1$ over all stretches tested, and each connected run of significant stretches is one dip.

b. Sweep ladder. SimpleSweep measures a uniform grid over the window with spacing Γ_0/p, $p = 2, 3, 4$, where Γ_0 is the FWHM of the prior-mean line, and repeats whole passes. Smart-Sweep sweeps at $\Gamma_0/3$ until the dip test finds a dip, then at $\Gamma_0/10$ inside the focus (below), and then measures only at the maxima of $g(x)^2$ at the posterior mean, cycling between them, once the posterior expects each of those points to carry at least 90% of the information of the true optimum.

c. Bayesian SBED. SBED is the particle filter choosing each x by Eq. (20). After every resampling it places 800 candidate drive frequencies at the quantiles of a density with Gaussian kernels on the flanks, $\pm$HWHM, of the posterior-mean line, kernels on the dips found and a flat share, thinned to a spacing of at least 100 kHz. Eq. (20) is evaluated on a 500-particle subsample, with σ^2 the posterior mean of the noise. The candidates are taken in blocks of 64, and the block maxima compete through a softmax with a temperature of 0.01 nats, so the choice is near-greedy. Candidates are restricted to the focus: one section per dip, its extent padded by one posterior-mean HWHM on each side. A new dip widens the focus at once; narrowing waits 8 measurements and must shrink it by at least 5%. The focus is re-decided after every resampling. Because a strategy that has piled its measurements onto a suspected dip cannot be contradicted there, SBED sometimes runs a falsification probe instead: with probability $0.02 + 0.08\,g$, where g is the widest unmeasured gap as a fraction of the window, it measures the middle of that gap. Smart-SBED is SBED that, until the first dip is found, picks the EIG maximiser among the unmeasured points of the $\Gamma_0/3$ grid instead, makes no probe and re-decides the focus after every measurement, re-running the dip test every 10 measurements.

C. Simulation setup and evaluation

a. What the claimed uncertainty is. The particle filter reports $\hat{\sigma}_N$, Eq. (29): a moment of the posterior, defined at every N. The van Trees bound, Eq. (16), limits the mean squared error averaged over the prior, and for the posterior mean that error equals the expected posterior variance, $\mathbb{E}[(\hat{\theta}_N - \theta)^2] = \mathbb{E}[\hat{\sigma}_N^2]$, when the posterior is correct. Comparing $\hat{\sigma}_N$ with the bound is therefore valid from the first measurement, as an average over runs and provided the posterior is calibrated. What fails before the posterior is Gaussian (Sec. II E) is the usual reading of $\hat{\sigma}_N$: with a uniform window or several live candidates it is not a 68% interval, and the posterior entropy is below $\frac{1}{2}\ln(2\pi e\,\hat{\sigma}_N^2)$, so information in bits, per-step gains as variance ratios, and coverage at Gaussian quantiles are only meaningful in the data regime. We therefore (i) compare $\hat{\sigma}_N$ with the bound as a run average at all N, (ii) test coverage, the true error against $\hat{\sigma}_N$, only once the posterior is Gaussian, or with quantiles of the particle distribution itself, and (iii) keep this separate from miscalibration of the particle filter, which makes $\hat{\sigma}_N$ too small at any N.

[Read every run against the bound: prior share J_0/J and information rate $(J(N) - J(N'))/((N - N')I^*)$, with $J_0 = \hat{\sigma}_0^{-2}$ of the initial cloud; a rate above 1 meaning the claimed σ fell faster than the data allow.]

[Use cases (SNR $\times$ window $\times$ Gaussian share $\times$ isotope), paired repeats, one shared budget (2500 measurements), coverage of the claimed σ, cost to reach τ; numbers per use case.]

IV. RESULTS

A. Simulations

[(a) Claimed σ and true error against Eqs. (16)–(18) for a narrow and a wide window. (b) Information rate and prior share. (c) Cost vs SNR and window. (d) Calibration of the claimed σ.]

B. Experiment

[Placeholder until the lab data are in.]

V. DISCUSSION

[Limits: the claimed σ is overconfident; the simulated prior is miscalibrated; Gaussian noise only; I^* assumes known nuisances; low-SNR detection model.]

VI. CONCLUSION

[Conclusion.]

[1] G. Balasubramanian, I. Y. Chan, R. Kolesov, M. Al-Hmoud, J. Tisler, C. Shin, C. Kim, A. Wojcik, P. R. Hemmer, A. Krueger, T. Hanke, A. Leitenstorfer, R. Bratschitsch, F. Jelezko, and J. Wrachtrup, Nature **455**, 648 (2008).

[2] J. R. Maze, P. L. Stanwix, J. S. Hodges, S. Hong, J. M. Taylor, P. Cappellaro, L. Jiang, M. V. G. Dutt, E. Togan, A. S. Zibrov, A. Yacoby, R. L. Walsworth, and M. D. Lukin, Nature **455**, 644 (2008).

[3] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, Physics Reports **528**, 1 (2013).

[4] S. Luo, Z. Zhu, L. Zhang, P. Chen, and R. Lin, Optical and Quantum Electronics **55**, 1004 (2023).

[5] G. Haim, C. Mullarkey, J. Howell, and N. Bar-Gill, Compressed sensing enabled high-bandwidth and large dynamic range magnetic sensing (2025), arXiv:2502.06070 [quant-ph].

[6] C. E. Granade, C. Ferrie, N. Wiebe, and D. G. Cory, New Journal of Physics **14**, 103013 (2012).

[7] C. Bonato, M. S. Blok, H. T. Dinani, D. W. Berry, M. L. Markham, D. J. Twitchen, and R. Hanson, Nature Nanotechnology **11**, 247 (2016).

[8] R. Santagati, A. A. Gentile, S. Knauer, S. Schmitt, S. Paesani, C. Granade, N. Wiebe, C. Osterkamp, L. P. McGuinness, J. Wang, M. G. Thompson, J. G. Rarity, F. Jelezko, and A. Laing, Physical Review X **9**, 021019 (2019).

[9] R. D. McMichael, S. Dushenko, and S. M. Blakley, Journal of Applied Physics **130**, 144401 (2021).

[10] S. Dushenko, K. Ambal, and R. D. McMichael, Physical Review Applied **14**, 054036 (2020).

[11] R. D. McMichael, S. M. Blakley, and S. Dushenko, Journal of Research of the National Institute of Standards and Technology **126**, 126002 (2021).

[12] A. Hentschel and B. C. Sanders, Physical Review Letters **104**, 063603 (2010).

[13] A. Lumino, E. Polino, A. S. Rab, G. Milani, N. Spagnolo, N. Wiebe, and F. Sciarrino, Physical Review Applied **10**, 044033 (2018).

[14] H. Xu, J. Li, L. Liu, Y. Wang, H. Yuan, and X. Wang, npj Quantum Information **5**, 82 (2019).

[15] J. Schuff, L. J. Fiderer, and D. Braun, New Journal of Physics **22**, 035001 (2020).

[16] T. Xiao, J. Fan, and G. Zeng, npj Quantum Information **8**, 2 (2022).

[17] V. Cimini, M. Valeri, E. Polino, S. Piacentini, F. Ceccarelli, G. Corrielli, N. Spagnolo, R. Osellame, and F. Sciarrino, Advanced Photonics **5**, 016005 (2023).

[18] L. J. Fiderer, J. Schuff, and D. Braun, PRX Quantum **2**, 020303 (2021).

[19] F. Belliardo, F. Zoratti, F. Marquardt, and V. Giovannetti, Quantum **8**, 1555 (2024).

[20] F. Belliardo, F. Zoratti, and V. Giovannetti, Physical Review A **109**, 062609 (2024).

[21] S. Guo, J. Liu, T. Le, and H. Dai, Scientific Reports 10.1038/s41598-026-76354-7 (2026).

[22] H. Cramér, Princeton University Press (1946).

[23] H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part I* (Wiley, New York, 1968).

[24] R. D. Gill and B. Y. Levit, Bernoulli **1**, 59 (1995).

[25] N. B. Manson, J. P. Harrison, and M. J. Sellars, Physical Review B **74**, 104303 (2006).

[26] S. Felton, A. M. Edmonds, M. E. Newton, P. M. Martineau, D. Fisher, D. J. Twitchen, and J. M. Baker, Physical Review B **79**, 075203 (2009).

[27] V. Jacques, P. Neumann, J. Beck, M. Markham, D. Twitchen, J. Meijer, F. Kaiser, G. Balasubramanian, F. Jelezko, and J. Wrachtrup, Physical Review Letters **102**, 057403 (2009).

[28] A. Dréau, M. Lesik, L. Rondin, P. Spinicelli, O. Arcizet, J.-F. Roch, and V. Jacques, Physical Review B **84**, 195204 (2011).

[29] A. W. van der Vaart, *Asymptotic Statistics* (Cambridge University Press, Cambridge, 1998).

[30] D. V. Lindley, The Annals of Mathematical Statistics **27**, 986 (1956).

[31] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, Hoboken, NJ, 2006).

[32] A. Doucet, N. de Freitas, and N. Gordon, eds., *Sequential Monte Carlo Methods in Practice* (Springer, New York, NY, 2001).

[33] G. Kitagawa, Journal of Computational and Graphical Statistics **5**, 1 (1996).

[34] J. Liu and M. West, in *Sequential Monte Carlo Methods in Practice*, edited by A. Doucet, N. de Freitas, and N. Gordon (Springer, New York, NY, 2001) pp. 197–223.